

1. Show the Algorithm in Action (Brain of the Tool)

Goal: Make panelists see *that Lazy HC2L and DHL both work and think*.

Must Have:

- Working **Lazy HC2L** (your proposed algorithm).
 - Displays how it computes and updates routes.
- Working **DHL** (baseline algorithm).
 - Used for direct comparison.
- A visible **toggle or dropdown** to choose between the two.
 - Label: `Select Algorithm: DHL | Lazy HC2L`

Show visually/logically:

LazyHC2L: `Impact = 0.45 <  $\tau(0.5)$  → Skipped update (lazy)`

DHL: `Immediate recomputation → 230 nodes updated`

 **This is your tool's “brain” — if this works, everything else builds around it.**

 **SOP 1:** Efficiency metrics

 **SOP 2:** Comparison responsiveness

2. Simulate One Clear Disruption Scenario

Goal: Prove that your tool adapts to road changes *intelligently*.

Must Have:

- “Add Disruption” button → click a road → mark road segment.
- Lazy HC2L decides:

- If **impact** < **threshold**, triggers lazy update.
- If **impact** ≥ **threshold**, skip the lazy update and proceed to immediate update.
- Updated nodes (affected area) highlighted in **green (Lazy)** or **yellow (DHL)**.
- “Reroute” button → shows new path instantly.

💡 Keep it simple:

- Scenario 1: Road closure on Katipunan → reroute through Commonwealth.
- Scenario 2: Light traffic → same route (no update).

✓ SOP 2: Responsiveness

✓ SOP 1: Performance under update

🎙️ *“See, it skipped update here because the disruption wasn’t severe, but when we closed Katipunan, it triggered an update and changed the route.”*

🧭 3. Route Comparison with Google Maps (Accuracy Proof)

Goal: Show your tool’s path is *similar* to a real-world route.

Must Have:

- Integrate **Google Maps Directions API** (for 1–2 routes only).
- Plot the **Google route (orange)** alongside:
 -  Lazy HC2L route
 -  DHL route
- Compute & display:
 - **Fréchet Distance (m)**

- **Segment Overlap (%)**

Example Output:

Algorithm: LazyHC2L

Fréchet Distance: 62.4 m

Segment Overlap: 89%

💡 *Even if done offline or pre-saved, you must display it visually.*

✅ SOP 3: Accuracy

✅ Also visually “completes” the tool — shows connection to real navigation.

4. Performance & Logging Dashboard

Goal: Make it *look scientific and measurable* (panel loves data).

Must Have:

- Side panel or bottom box showing:
 - Labeling Time (s)
 - Labeling Size (MB)
 - Query Response Time (μs)
 - Nodes Updated
 - Fréchet Distance
 - Segment Overlap

Export all metrics to CSV:

```
trial, algorithm, label_time, label_size, query_time, frechet,
overlap, nodes_updated
```

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Example Visual:

Metric	DHL	LazyHC2L
Label Time (s)	3.4	2.8
Label Size (MB)	58	47
Query (μ s)	130	110

- ✓ SOP 1: Efficiency
- ✓ SOP 2: Responsiveness
- ✓ SOP 3: Accuracy

💡 Even a simple HTML table counts as a “dashboard.”

5. Clean, Understandable UI

Goal: Make it look *complete* and easy to follow during your presentation.

Must Have:

- **Base map** (Leaflet.js + OSMnx)
- **Color legend:**
 -  Lazy HC2L
 -  DHL
 -  Google
 -  Disruption
 -  Lazy update region
 -  DHL update region

Log panel (or console output):

Impact: $0.64 \geq \text{Threshold } 0.5 \rightarrow \text{Triggered Lazy Update}$
Updated nodes: [34–40]

Query time: 115 μ s

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- **Buttons:**
 - “Add Disruption”
 - “Reroute”
 - “Reset”
 - “Export Results”

✓ *SOP visual support for all three.*

💡 Simple and clear beats fancy and confusing.

6. Single Click Demo Scenario

Goal: Run your whole defense demo in under 2 minutes smoothly.

Must Have:

- “▶ Run Demo” button or ready-made sequence:
 1. Show normal route (Lazy HC2L)
 2. Add disruption
 3. Lazy update triggers → reroute shown
 4. Google route overlay appears → compare similarity
 5. Dashboard updates metrics

✓ *Makes your defense seamless and structured.*

FINAL MUST-HAVE LIST (Non-Negotiable for Demo)

Priority	Feature	Purpose	SOP
 1	Working Lazy HC2L + DHL toggle	Core comparison	1 & 2
 2	Disruption simulation + reroute	Show responsiveness	2
 3	Google Maps route + metrics	Accuracy proof	3
 4	Dashboard (time, memory, overlap, etc.)	Data visualization	1–3
 5	Clean map UI + legend	Understandable presentation	all
 6	Demo button or flow	Smooth presentation	all